

B1 Fluids, 2025 — Model Solutions

Question 1 — Streamfunction, vorticity and point vortices

Problem statement. Introduce the streamfunction ψ for 2D incompressible flow and prove the Bernoulli–vorticity relations $\omega = \omega(\psi)$, $B = B(\psi)$, $\omega = -dB/d\psi$. Apply to a Rankine vortex, and to two point vortices of strengths A and B .

(a) Continuity, streamfunction and vorticity

Continuity for a compressible fluid:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (1)$$

Incompressibility ($D\rho/Dt = 0$) reduces this to

$$\nabla \cdot \mathbf{v} = 0.$$

In 2D, $\partial_x v_x + \partial_y v_y = 0$ is the integrability condition for $\psi(x, y)$ with

$$v_x = -\frac{\partial \psi}{\partial y}, \quad v_y = \frac{\partial \psi}{\partial x},$$

i.e. $\mathbf{v} = -\hat{\mathbf{z}} \times \nabla \psi$. The vorticity $\omega = \omega \hat{\mathbf{z}}$ has

$$\omega = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = \nabla^2 \psi. \quad (2)$$

Lines of constant ψ are streamlines since $\mathbf{v} \cdot \nabla \psi = 0$.

(b) Bernoulli–vorticity theorem for steady 2D inviscid flow

Euler:

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p - g \hat{\mathbf{z}}. \quad (3)$$

Using $(\mathbf{v} \cdot \nabla) \mathbf{v} = \nabla(\frac{1}{2}v^2) - \mathbf{v} \times (\nabla \times \mathbf{v})$, steady flow in a horizontal plane (gravity absorbed into p):

$$\nabla \left(\frac{p}{\rho} + \frac{v^2}{2} \right) = \mathbf{v} \times (\omega \hat{\mathbf{z}}) = \omega \nabla \psi,$$

since $\mathbf{v} \times \hat{\mathbf{z}} = \nabla \psi$. With $B \equiv p/\rho + \frac{1}{2}v^2$,

$$\nabla B = \omega \nabla \psi. \quad (4)$$

Dotting with $\mathbf{v} \perp \nabla \psi$ gives $\mathbf{v} \cdot \nabla B = 0$, so

$$B = B(\psi). \quad (5)$$

Curl of (3) kills both gradient terms, so $D\omega/Dt = 0$; in steady flow,

$$\boxed{\omega = \omega(\psi)}. \quad (6)$$

Substituting $\nabla B = (dB/d\psi)\nabla\psi$ in (4) and fixing the sign by the rigid-rotation check ($\psi = -\frac{1}{2}\Omega r^2$, $\omega = 2\Omega$):

$$\boxed{\omega = -\frac{dB}{d\psi}}. \quad (7)$$

(c) Rankine vortex

Symmetry gives $\mathbf{v} = v_\theta(r)\hat{\boldsymbol{\theta}}$ and

$$\omega = \frac{1}{r} \frac{d}{dr}(rv_\theta).$$

Inside ($r \leq r_0$, $\omega = \omega_0$), regularity at $r = 0$:

$$v_\theta = \frac{1}{2}\omega_0 r.$$

Outside ($\omega = 0$), $rv_\theta = \text{const}$; matching at r_0 :

$$\boxed{v_\theta(r) = \begin{cases} \frac{1}{2}\omega_0 r, & r \leq r_0, \\ \frac{1}{2}\omega_0 r_0^2/r, & r > r_0. \end{cases}} \quad (8)$$

Radial Euler (centripetal balance):

$$\frac{1}{\rho} \frac{dp}{dr} = \frac{v_\theta^2}{r}.$$

Outside, substitute $v_\theta = \omega_0 r_0^2/(2r)$:

$$\frac{v_\theta^2}{r} = \frac{\omega_0^2 r_0^4}{4r^3}.$$

Integrate from r to ∞ with $p(\infty) = p_\infty$:

$$p_\infty - p(r) = \int_r^\infty \frac{\rho\omega_0^2 r_0^4}{4r'^3} dr' = \frac{\rho\omega_0^2 r_0^4}{8r^2}.$$

Hence outside,

$$p(r) = p_\infty - \frac{\rho\omega_0^2 r_0^4}{8r^2}.$$

Inside, substitute $v_\theta = \frac{1}{2}\omega_0 r$:

$$\frac{v_\theta^2}{r} = \frac{1}{4}\omega_0^2 r.$$

Integrate:

$$p(r) = \frac{1}{8}\rho\omega_0^2 r^2 + C.$$

Apply continuity at $r = r_0$:

$$\frac{1}{8}\rho\omega_0^2 r_0^2 + C = p_\infty - \frac{1}{8}\rho\omega_0^2 r_0^2.$$

Solve for C :

$$C = p_\infty - \frac{1}{4}\rho\omega_0^2 r_0^2.$$

Therefore

$$\boxed{p(r) = \begin{cases} p_\infty - \frac{1}{4}\rho\omega_0^2 r_0^2 + \frac{1}{8}\rho\omega_0^2 r^2, & r \leq r_0, \\ p_\infty - \frac{\rho\omega_0^2 r_0^4}{8r^2}, & r > r_0. \end{cases}} \quad (9)$$

(d) Two point vortices

Each vortex moves only in the field of the other. With separation $\mathbf{d} = \mathbf{x}_2 - \mathbf{x}_1$, $L = |\mathbf{d}|$:

$$\dot{\mathbf{x}}_1 = \frac{B}{2\pi L^2} \hat{\mathbf{z}} \times \mathbf{d}, \quad \dot{\mathbf{x}}_2 = -\frac{A}{2\pi L^2} \hat{\mathbf{z}} \times \mathbf{d}. \quad (10)$$

Hence

$$\dot{\mathbf{d}} = -\frac{A+B}{2\pi L^2} \hat{\mathbf{z}} \times \mathbf{d}, \quad A\mathbf{x}_1 + B\mathbf{x}_2 = \text{const.} \quad (11)$$

Since $\dot{\mathbf{d}} \perp \mathbf{d}$, L is constant.

Case $A = B$. Centroid fixed; both vortices co-rotate about the midpoint on circles of radius $L/2$.

Case $A = -B$. $\dot{\mathbf{d}} = 0$; (11) gives equal translational velocity $\dot{\mathbf{x}}_1 = \dot{\mathbf{x}}_2 = (B/2\pi L^2) \hat{\mathbf{z}} \times \mathbf{d}$. Pair propagates as two parallel straight lines.

Sketch.

$A = B$ (co-rotating)

$A = -B$ (translating)



Frequency (periodic case $A = B$). From (11), $|\dot{\mathbf{d}}| = \Omega L$ with $\Omega = (A+B)/(2\pi L^2)$:

$$\boxed{\omega_{\text{orb}} = \frac{A+B}{2\pi L^2} = \frac{A}{\pi L^2} \quad (A = B).} \quad (12)$$

Question 2 — Complex potential and conformal mappings

Problem statement. For analytic $w(z) = \phi + i\psi$, derive Cauchy–Riemann, harmonicity and orthogonality of level curves. Apply to $w = Bz^2$. Define a conformal mapping; invert $Z = z/2 + \sqrt{z^2/4 - c^2}$ and find Φ, Ψ for $w(z) = Az$ in the Z -plane.

(a) Analytic functions and Cauchy–Riemann

Analytic w : $w'(z)$ direction-independent. Equating $\Delta z = \Delta x$ and $\Delta z = i\Delta y$ limits,

$$w'(z) = \frac{\partial \phi}{\partial x} + i \frac{\partial \psi}{\partial x} = -i \frac{\partial \phi}{\partial y} + \frac{\partial \psi}{\partial y}.$$

Real/imaginary parts:

$$\boxed{\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}, \quad \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}.} \quad (13)$$

Cross-differentiating (13),

$$\nabla^2 \phi = 0, \quad \nabla^2 \psi = 0.$$

Orthogonality:

$$\nabla \phi \cdot \nabla \psi = (\partial_y \psi)(\partial_x \phi) + (-\partial_x \psi)(\partial_y \phi) = 0.$$

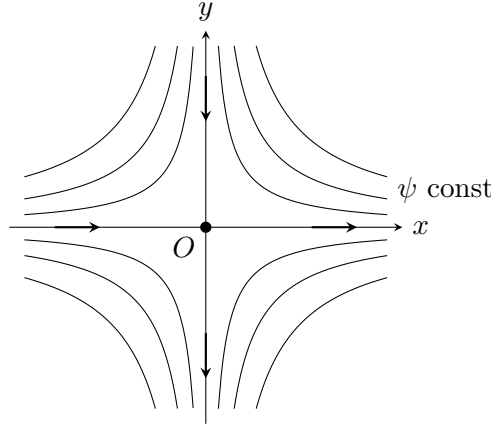
For 2D incompressible irrotational inviscid flow, ϕ is the velocity potential ($\mathbf{v} = \nabla \phi$) and ψ the streamfunction; (13) reproduces $v_x = \partial_x \phi = \partial_y \psi$, $v_y = \partial_y \phi = -\partial_x \psi$.

(b) The flow $w(z) = Bz^2$

$w = B(x^2 - y^2) + 2iBxy$, so

$$\boxed{\phi = B(x^2 - y^2), \quad \psi = 2Bxy.} \quad (14)$$

Streamlines $\psi = \text{const}$ are hyperbolae $xy = \text{const}$ ($\psi = 0$ is the axes); equipotentials $x^2 - y^2 = \text{const}$ are the orthogonal family.



For $B > 0$: in the first quadrant fluid descends along the $+y$ axis and exits along the $+x$ axis.

Velocity: $u - iv = w'(z) = 2Bz$, so $\mathbf{v} = (2Bx, -2By)$. The point $w'(z) = 0$ at $z = 0$ is a stagnation point: 2D corner-flow saddle, fluid against a right-angled wall.

(c) Conformal mapping and Joukowski plate

A holomorphic $Z = f(z)$ with $f' \neq 0$ is a conformal map: angles preserved. If $w(z)$ is analytic then $W(Z) \equiv w(f(z))$ with $z = f^{-1}(Z)$ is analytic in Z and represents another potential flow; streamlines map by f .

Invert $Z = z/2 + \sqrt{z^2/4 - c^2}$. Subtract $z/2$:

$$Z - \frac{z}{2} = \sqrt{\frac{z^2}{4} - c^2}.$$

Square both sides:

$$\left(Z - \frac{z}{2}\right)^2 = \frac{z^2}{4} - c^2.$$

Expand the left-hand side:

$$Z^2 - zZ + \frac{z^2}{4} = \frac{z^2}{4} - c^2.$$

Cancel $z^2/4$:

$$Z^2 - zZ = -c^2.$$

Solve for z :

$$zZ = Z^2 + c^2.$$

Divide by Z :

$$z = F(Z) = Z + \frac{c^2}{Z}. \quad (15)$$

On $|Z| = c$, write $Z = ce^{i\theta}$:

$$z = ce^{i\theta} + ce^{-i\theta} = 2c \cos \theta \in [-2c, 2c].$$

For $w(z) = Az$ (uniform stream of speed A),

$$W(Z) = A \left(Z + \frac{c^2}{Z} \right).$$

With $Z = Re^{i\Theta}$,

$$\Phi = A \left(R + \frac{c^2}{R} \right) \cos \Theta, \quad \Psi = A \left(R - \frac{c^2}{R} \right) \sin \Theta. \quad (16)$$

$\Psi = 0$ on the real axis or on $R = c$: (16) is uniform flow past a rigid cylinder of radius c . In the z -plane the cylinder collapses to the flat plate $z \in [-2c, 2c]$, so $w = Az$ in z is uniform flow past a horizontal flat plate of length $4c$.

Real fluid. Viscosity enforces no-slip, generating a boundary layer that separates on the rear half at large Re , producing a wake and finite drag (resolving d'Alembert's paradox). The wake is unsteady (von Kármán street), so the symmetric streamlines of (16) are never observed near the body; the conformal result describes only the outer irrotational flow.

Question 3 — Thin viscous film on an inclined conveyor belt

Problem statement. A thin viscous film $h(x, t)$ on a belt inclined at θ , moving at $U_0 \hat{\mathbf{x}}$; y normal to belt. Define Re , derive the lubrication equation, find $v_x(y)$, the equilibrium thickness, and timescales after the belt stops.

(a) Reynolds number and lubrication

$$\text{Re} = \frac{|(\mathbf{v} \cdot \nabla) \mathbf{v}|}{|\nu \nabla^2 \mathbf{v}|} \sim \frac{UL}{\nu}.$$

$\text{Re} \ll 1$: viscous diffusion dominates (Stokes). For a thin film, aspect ratio $\epsilon = h/L \ll 1$; incompressibility gives $v_y \sim \epsilon U_0$, and the inertial term is smaller than $\nu \partial_y^2 v_x$ by $\epsilon \text{Re} \ll 1$:

$$0 = -\frac{1}{\rho} \frac{\partial p}{\partial x} + g \sin \theta + \nu \frac{\partial^2 v_x}{\partial y^2}, \quad 0 = -\frac{1}{\rho} \frac{\partial p}{\partial y} - g \cos \theta. \quad (17)$$

Streamwise viscous term $\nu \partial_x^2 v_x$ suppressed by ϵ^2 .

Integrate (17)₂ in y :

$$p(x, y) = -\rho g \cos \theta y + f(x).$$

Apply $p(y = h) = p_a$ (no surface tension):

$$f(x) = p_a + \rho g \cos \theta h(x).$$

Hence

$$p = p_a + \rho g \cos \theta (h - y).$$

Differentiate with respect to x :

$$\frac{\partial p}{\partial x} = \rho g \cos \theta \frac{\partial h}{\partial x}.$$

Substitute into (17)₁:

$$\boxed{g \frac{\partial h}{\partial x} \cos \theta + g \sin \theta = \nu \frac{\partial^2 v_x}{\partial y^2}.} \quad (18)$$

(b) Boundary conditions and velocity profile

No-slip at belt; no tangential stress at the free surface:

$$v_x(0) = U_0, \quad \left. \frac{\partial v_x}{\partial y} \right|_{y=h} = 0.$$

Write $G \equiv g \cos \theta \partial h / \partial x + g \sin \theta$, so (18) reads $\nu \partial^2 v_x / \partial y^2 = G$. Integrate once in y :

$$\nu \frac{\partial v_x}{\partial y} = Gy + C_1.$$

Apply $\partial_y v_x|_{y=h} = 0$:

$$C_1 = -Gh.$$

So

$$\nu \frac{\partial v_x}{\partial y} = G(y - h).$$

Integrate again:

$$\nu v_x = G\left(\frac{1}{2}y^2 - hy\right) + C_2.$$

Apply $v_x(0) = U_0$:

$$C_2 = \nu U_0.$$

Therefore

$$\boxed{v_x(x, y, t) = U_0 + \frac{1}{\nu} \left[g \cos \theta \frac{\partial h}{\partial x} + g \sin \theta \right] \left(\frac{1}{2}y^2 - hy \right).} \quad (19)$$

(c) Equilibrium thickness

Steady, $\partial h / \partial x = 0$:

$$v_x(y) = U_0 + \frac{g \sin \theta}{\nu} \left(\frac{1}{2}y^2 - h_{\text{eq}}y \right). \quad (20)$$

Flux per unit width:

$$Q = \int_0^{h_{\text{eq}}} v_x \, dy.$$

Substitute (20):

$$Q = \int_0^{h_{\text{eq}}} \left[U_0 + \frac{g \sin \theta}{\nu} \left(\frac{1}{2}y^2 - h_{\text{eq}}y \right) \right] dy.$$

Integrate term by term:

$$Q = U_0 h_{\text{eq}} + \frac{g \sin \theta}{\nu} \left(\frac{1}{6} h_{\text{eq}}^3 - \frac{1}{2} h_{\text{eq}}^3 \right).$$

Simplify:

$$Q = U_0 h_{\text{eq}} - \frac{g \sin \theta h_{\text{eq}}^3}{3\nu}.$$

Impose no net transport, $Q = 0$:

$$U_0 = \frac{g \sin \theta h_{\text{eq}}^2}{3\nu}.$$

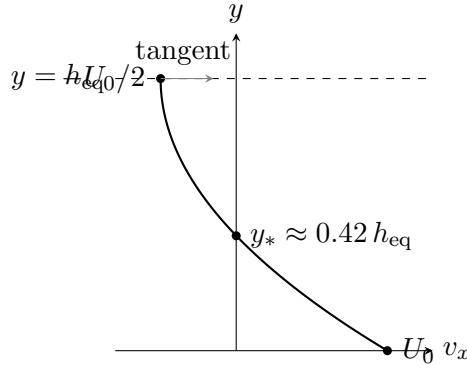
Solve for h_{eq} :

$$h_{\text{eq}} = \sqrt{\frac{3\nu U_0}{g \sin \theta}}. \quad (21)$$

At $y = h_{\text{eq}}$, using $g \sin \theta h_{\text{eq}}^2 / \nu = 3U_0$:

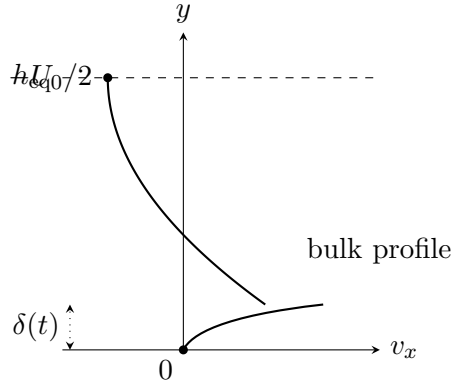
$$v_x(h_{\text{eq}}) = U_0 + 3U_0\left(\frac{1}{2} - 1\right) = U_0 - \frac{3}{2}U_0 = -\frac{1}{2}U_0.$$

Zero-crossing $v_x = 0$ at $y_* = h_{\text{eq}}(1 - 1/\sqrt{3}) \approx 0.42 h_{\text{eq}}$.



(d) Belt switched off

Just after stop, no-slip gives $v_x(0) = 0$; momentum redistributes via a thin viscous layer $\delta(t) \sim \sqrt{\nu t}$ at the wall while the bulk retains (20).



Sketch (i): equilibrium parabola with a thin viscous kink near $y = 0$ pulling v_x to 0.

Once $\delta \sim h_{\text{eq}}$, the wall layer fills the film. Integrate $\nu \partial_y^2 v_x = -g \sin \theta$ once:

$$\nu \frac{\partial v_x}{\partial y} = -g \sin \theta y + C_1.$$

Apply $\partial_y v_x(h) = 0$:

$$C_1 = g \sin \theta h.$$

Integrate again:

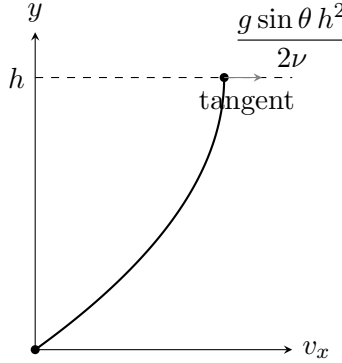
$$\nu v_x = g \sin \theta \left(hy - \frac{1}{2} y^2 \right) + C_2.$$

Apply $v_x(0) = 0$:

$$C_2 = 0.$$

Hence

$$v_x(y) = \frac{g \sin \theta}{\nu} \left(hy - \frac{1}{2} y^2 \right). \quad (22)$$



Sketch (ii): drainage parabola from 0 at $y = 0$ to maximum $g \sin \theta h^2 / (2\nu)$ at $y = h$, with horizontal tangent at the free surface.

Diffusion timescale. $\partial_t v_x = \nu \partial_y^2 v_x$ across h_{eq} :

$$\boxed{\tau_{\text{diff}} \sim \frac{h_{\text{eq}}^2}{\nu}.} \quad (23)$$

Drainage timescale. Surface speed $U_{\text{drain}} \sim g \sin \theta h_{\text{eq}}^2 / \nu$; using (21), $g \sin \theta h_{\text{eq}}^2 = 3\nu U_0$:

$$\boxed{\tau_{\text{drain}} \sim \frac{L\nu}{g \sin \theta h_{\text{eq}}^2} \sim \frac{L}{U_0}.} \quad (24)$$

$\tau_{\text{drain}} / \tau_{\text{diff}} \sim L / h_{\text{eq}} \gg 1$ (lubrication limit): fast diffusion sets up the drainage profile, slow drainage empties the film.

Question 4 — Stratified instability and Rayleigh–Bénard convection

Problem statement. (a) Derive the buoyancy frequency for incompressible $\rho(z)$ and discuss instability. (b) Derive the perturbation vorticity equation and the sixth-order equation for v'_z in Boussinesq Rayleigh–Bénard. (c) For the mode $v'_z = A \sin(kx) \sin(\pi z/H) e^{\sigma t}$, find σ and the marginal Rayleigh number.

(a) Brunt–Väisälä frequency

A parcel displaced by ζ keeps density $\rho(z_0)$; surroundings have $\rho(z_0 + \zeta) \approx \rho(z_0) + \zeta \, d\rho/dz$. Buoyancy force per unit volume:

$$F_b = g\zeta \frac{d\rho}{dz}.$$

Newton, $\rho(z_0)\ddot{\zeta} = F_b$, gives $\ddot{\zeta} = -\omega^2\zeta$ with

$$\boxed{\omega^2 = -\frac{g}{\rho} \frac{d\rho}{dz}}. \quad (25)$$

Stable ($d\rho/dz < 0$): oscillation. Unstable ($d\rho/dz > 0$): $\omega^2 < 0$, exponential growth $\zeta \propto e^{|\omega|t}$ (Rayleigh–Taylor).

(b) Equilibrium and perturbation equations

Equilibrium: $\kappa \nabla^2 T = 0$, so

$$\boxed{T_{\text{eq}}(z) = T_0 - \frac{\Delta T}{H} z}. \quad (26)$$

With $b = g\alpha(T - T_0)$, hydrostatic balance gives $p_{\text{eq}}(z) = p_0 - \rho_0 g\alpha \Delta T z^2 / (2H)$.

Linearising $\mathbf{v} = (v'_x, v'_z)$, $T = T_{\text{eq}} + T'$, $p = p_{\text{eq}} + p'$:

$$\frac{\partial v'_x}{\partial t} + \frac{1}{\rho_0} \frac{\partial p'}{\partial x} = \nu \nabla^2 v'_x, \quad (27)$$

$$\frac{\partial v'_z}{\partial t} - g\alpha T' + \frac{1}{\rho_0} \frac{\partial p'}{\partial z} = \nu \nabla^2 v'_z, \quad (28)$$

$$\frac{\partial T'}{\partial t} - \frac{\Delta T}{H} v'_z = \kappa \nabla^2 T', \quad (29)$$

$$\frac{\partial v'_x}{\partial x} + \frac{\partial v'_z}{\partial z} = 0. \quad (30)$$

$\partial_z(27) - \partial_x(28)$ eliminates p' :

$$\boxed{\left(\frac{\partial}{\partial t} - \nu \nabla^2\right) \left(\frac{\partial v'_x}{\partial z} - \frac{\partial v'_z}{\partial x}\right) = -g\alpha \frac{\partial T'}{\partial x}}. \quad (31)$$

Apply ∂_x and use (30) to write $\partial_x(\partial_z v'_x - \partial_x v'_z) = -\nabla^2 v'_z$:

$$\left(\frac{\partial}{\partial t} - \nu \nabla^2\right) \nabla^2 v'_z = g\alpha \frac{\partial^2 T'}{\partial x^2}.$$

Apply $(\partial_t - \kappa \nabla^2)$ and use (29) as $(\partial_t - \kappa \nabla^2)T' = (\Delta T/H)v'_z$:

$$\boxed{\left(\frac{\partial}{\partial t} - \nu \nabla^2\right) \left(\frac{\partial}{\partial t} - \kappa \nabla^2\right) \nabla^2 v'_z = \frac{g\alpha \Delta T}{H} \frac{\partial^2 v'_z}{\partial x^2}}. \quad (32)$$

(c) Normal modes and Rayleigh criterion

For $v'_z = A \sin(kx) \sin(\pi z/H) e^{\sigma t}$, $\nabla^2 \rightarrow -K^2$ with $K^2 = k^2 + (\pi/H)^2$, $\partial_t \rightarrow \sigma$, $\partial_x^2 \rightarrow -k^2$. Substitute into (32):

$$(\sigma + \nu K^2)(\sigma + \kappa K^2)(-K^2) v'_z = -\frac{g\alpha\Delta T}{H} k^2 v'_z.$$

Cancel v'_z and a factor of -1 :

$$(\sigma + \nu K^2)(\sigma + \kappa K^2) K^2 = \frac{g\alpha\Delta T}{H} k^2.$$

Divide by K^2 :

$$(\sigma + \nu K^2)(\sigma + \kappa K^2) = \frac{g\alpha\Delta T}{H} \frac{k^2}{K^2}.$$

Expand as a quadratic in σ :

$$\sigma^2 + (\nu + \kappa) K^2 \sigma + \nu \kappa K^4 - \frac{g\alpha\Delta T}{H} \frac{k^2}{K^2} = 0.$$

Solve:

$$\sigma = -\frac{1}{2}(\nu + \kappa) K^2 \pm \sqrt{\frac{1}{4}(\nu - \kappa)^2 K^4 + \frac{g\alpha\Delta T}{H} \frac{k^2}{K^2}}. \quad (33)$$

Instability ($\sigma > 0$) requires the constant term of the quadratic to be negative:

$$\nu \kappa K^4 - \frac{g\alpha\Delta T}{H} \frac{k^2}{K^2} < 0.$$

Rearrange:

$$\frac{g\alpha\Delta T}{\nu \kappa H} > \frac{K^6}{k^2}. \quad (34)$$

Minimise $f(k) = (k^2 + \pi^2/H^2)^3/k^2$. Differentiate:

$$\frac{df}{dk} = \frac{3(k^2 + \pi^2/H^2)^2 \cdot 2k \cdot k^2 - (k^2 + \pi^2/H^2)^3 \cdot 2k}{k^4}.$$

Factor:

$$\frac{df}{dk} = \frac{2(k^2 + \pi^2/H^2)^2}{k^3} [3k^2 - (k^2 + \pi^2/H^2)].$$

Set bracket to zero:

$$3k^2 = k^2 + \pi^2/H^2.$$

Solve:

$$k_c^2 = \frac{\pi^2}{2H^2}.$$

Hence

$$K_c^2 = k_c^2 + \frac{\pi^2}{H^2} = \frac{3\pi^2}{2H^2}.$$

Compute the threshold:

$$\frac{K_c^6}{k_c^2} = \frac{(3\pi^2/2H^2)^3}{\pi^2/2H^2} = \frac{27\pi^6/8H^6}{\pi^2/2H^2} = \frac{27\pi^4}{4H^4}.$$

Hence

$$\text{Ra} \equiv \frac{g\alpha\Delta T H^3}{\nu \kappa} > \frac{27\pi^4}{4} \approx 657.5. \quad (35)$$

This applies to stress-free isothermal boundaries (the sine ansatz). Rigid no-slip raises $\text{Ra}_c \approx 1708$. Critical wavelength $\lambda_c = 2\pi/k_c = 2\sqrt{2}H$: rolls of horizontal extent $\sim H$. At threshold $\sigma = 0$ (steady bifurcation).